baseline), and improve how I gather this data in the future to make better decisions. Like most people, I find being able to place data in its spatial context helps me to understand it, and so I will need a model of my home that will show the data in context, as well as a floorplan to map out future sensor placement.

### 2.2.2 Start with what you have

I know that I will need to add a range of sensors to my home to gather the key data about the physical environment that will be required to support my objectives. But before I rush out and buy sensors, I can use a range of information sources that I already have to build my digital representation and to establish a digital baseline of the metrics of interest. This is just as true in many organizations looking to build a digital twin as it is in my example. The physical system you are building a twin of is likely already captured in many sources that you can use to build your digital representation. Lets look at what these sources of information are—not all of them will be relevant for my home digital twin, but as you look to larger and more complex systems, they will be increasingly important.

## 2.3 Information sources for digital representation

Most physical systems are extensively documented via drawings, maintenance logs, reports, and photos. This multi-modal documentation holds decades of operational knowledge, forming the foundation of current asset management.

However, much of this information is in formats digital twin platforms can’t directly process, for example scanned PDFs, paper records, and legacy files. Transforming these diverse sources into machine-readable formats is an important first step in building comprehensive digital representations without starting from scratch.

### 2.3.1 Historical records

Text-based documents capturing a system’s state over time are essential for digital representation, providing data for change analysis and machine